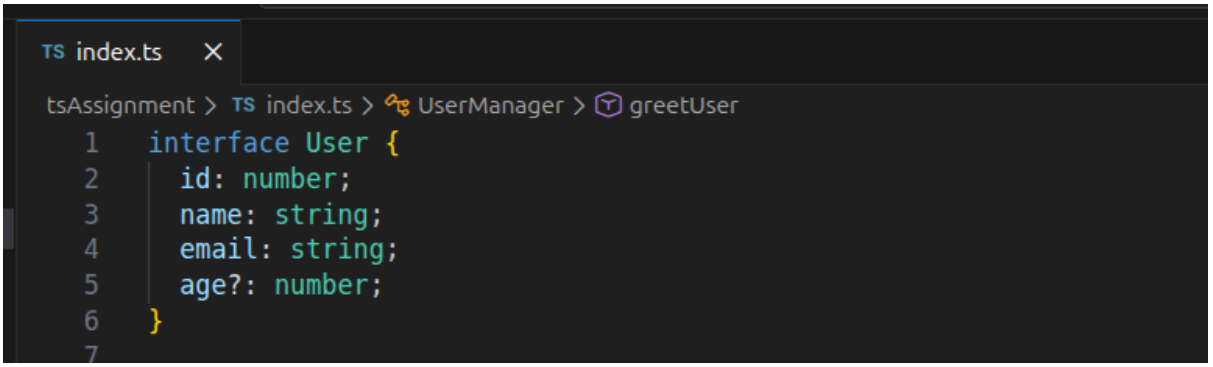
TS Assignment

Q1 Define an interface User with the following properties:
○ id(number), name(string), email(string), age(number optional)



Q2 Create a class UserManager with:
○ A private array users: User[] to store user data.
○ A method addUser(user: User): void that adds a new user.
○ A method removeUser(id: number): void that removes a user by ID.
○ A method getUser(id: number): User | undefined that retrieves a user by ID.
○ A method getAllUsers(): User[] that returns all users.

Code -

```
interface User {
 id: number;
 name: string;
 email: string;
 age?: number;
}

class UserManager {
 private users: User[] = [];

 addUser(user: User): void {
   this.users.push(user);
 }

 removeUser(id: number): void {
   this.users = this.users.filter((user) => user.id !== id);
 }

 getUser(id: number): User | undefined {
   for (let user of this.users) {
```

```
    if (user.id === id) {
      return user;
    }
  }
}

getAllUsers(): User[] {
  return this.users;
}

greetUser(name: string = "Guest"): string {
  const user = this.users.find((u) => u.name === name);
  return user ? `Hello ${user.name}` : "Hello Guest";
}

printUserDetails(targetUser: User): void {
  for (let user of this.users) {
      if (user.id === targetUser.id) {
          const { id, name, email, age } = user;
          console.log(id, name, email, age);
          return;
      }
  }
  console.log("User not found");
}

}

const obj1 : User = {
 id: 5,
 name: "aman",
 email: "aman@gmail.com",
 age: 20,
};

const obj2 : User = {
 id: 10,
 name: "dhruv",
 email: "dhruv@gmail.com",
 age: 22,
};
```

```typescript
const obj3 : User = {
 id: 15,
 name: "Harsh",
 email: "harsh@gmail.com",
 age: 25,
};
const person1 = new UserManager();
person1.addUser(obj1);
person1.addUser(obj2);
person1.addUser(obj3);
console.log(person1.getAllUsers());

console.log(person1.getUser(10));
person1.removeUser(10);

console.log(person1.getAllUsers());

console.log(person1.greetUser("fdf"));
console.log(person1.greetUser("aman"));

person1.printUserDetails(obj1)
```

```
[Running] node "/home/aman-mittal/Desktop/typescript/tsAssignment/index.js"
[
  { id: 5, name: 'aman', email: 'aman@gmail.com', age: 20 },
  { id: 10, name: 'dhruv', email: 'dhruv@gmail.com', age: 22 },
  { id: 15, name: 'Harsh', email: 'harsh@gmail.com', age: 25 }
]
{ id: 10, name: 'dhruv', email: 'dhruv@gmail.com', age: 22 }
[
  { id: 5, name: 'aman', email: 'aman@gmail.com', age: 20 },
  { id: 15, name: 'Harsh', email: 'harsh@gmail.com', age: 25 }
]
```

Q3 Use Arrow Functions & Default Parameters

○ Add a method greetUser = (name: string = "Guest"): string that returns a greeting message.

```
  greetUser(name: string = "Guest"): string {
    const user = this.users.find((u) => u.name === name);
    return user ? `Hello ${user.name}` : "Hello Guest";
  }
```

Q4 Use Destructuring & Spread Operator

○ Create a function printUserDetails(user: User): void that logs user details using object Destructuring.

```
  printUserDetails(targetUser: User): void {
    for (let user of this.users) {
        if (user.id === targetUser.id) {
            const { id, name, email, age } = user;
            console.log(id, name, email, age);
            return;
        }
    }
    console.log("User not found");
}
```